SARATHKUMAR DUNNAPAPAGARI

Cloud & Software Engineer · M.Tech CSE Student

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PROFILE

M.Tech Computer Science student and B.Tech IT graduate with four internships and trainings across AWS cloud, Python development and data science / machine learning. I have built real-time computer vision, ML and full-stack projects with Python, PyTorch, YOLOv8, FastAPI and React, and I deploy on AWS and GCP. Looking for a full-time role in cloud and software engineering where I can turn ideas into reliable, scalable solutions.

TECHNICAL SKILLS

Cloud	AWS (EC2, S3, IAM), Google Cloud Platform (GCP)
Languages	Python, Java, C
ML & Data	PyTorch, scikit-learn, YOLOv8, OpenCV, Pandas, DuckDB, Computer Vision
Web & APIs	FastAPI, React, HTML, CSS, JavaScript
DevOps & Tools	Docker, Git & GitHub, Prometheus, Grafana, Linux, Unix, Windows, MS Office, AutoCAD

PROJECTS

Construction Site Safety Monitor GitHub · Project page **Featured**

- Real-time video pipeline for webcam, CCTV (RTSP) or recorded footage that runs YOLOv8 detection and streams annotated frames and live detections to a React dashboard over WebSocket.
- PPE-ready: helmet / vest violation alerts with S3 snapshots activate once a custom PPE model is configured.
- Containerized with Docker Compose and NGINX, with an EC2 setup script and Prometheus + Grafana monitoring.

Tech: Python, YOLOv8, OpenCV, FastAPI, React + TypeScript, Docker, AWS S3, Prometheus, Grafana

Data Analyst + ML Agent GitHub **Full-stack**

- Platform for AI-assisted data analysis: upload a dataset, profile and clean it, explore it with charts, plain-English questions or SQL, then train, compare and explain ML models.
- Exports PDF reports; optional LLM explanations layer on code-computed results, and it works without an API key.

Tech: FastAPI, Pandas, DuckDB, scikit-learn, React, TypeScript, Tailwind CSS

EmoMusic GitHub **Computer Vision**

- Recommends music from real-time facial emotion recognition, building a playlist that gradually moves the listener toward a calmer, more positive mood.
- Fine-tuned a transfer-learned CNN on FER-2013, with smoothed predictions for a stable mood and playlist.

Tech: PyTorch, timm, OpenCV

EDUCATION

M.Tech, Computer Science and Engineering — GMR Institute of Technology, Rajam · *Pursuing* **2025 – 2027**

B.Tech, Information Technology — Sir C.R. Reddy College of Engineering, Eluru · **CGPA 7.29** **2021 – 2024**

Diploma, Civil Engineering — M.R.A.G.R Government Polytechnic, Vizianagaram · **67.69%** **2017 – 2020**

INTERNSHIPS

AWS Cloud Intern — BrainOVision Solutions **2 months**

- Deployed applications on AWS EC2, managed S3 storage and explored IAM roles.
- Gained hands-on experience with cloud infrastructure setup and monitoring.

Data Science & Machine Learning Intern — APSCHE **6 months**

- Worked on predictive modeling and AI-based solutions for real-world applications.
- Applied ML algorithms to datasets and improved accuracy through data preprocessing.

Python Developer Intern — BrainOVision Solutions **2 months**

- Developed automation scripts and small-scale applications in Python.
- Improved program efficiency with optimized coding practices.

Industrial Training — APSPHC **6 months**

- Gained exposure to infrastructure and technical workflows in public sector projects.

CERTIFICATIONS

- **Google Cloud Computing Foundations** (NPTEL) · **Cloud Computing** (NPTEL) · **Introduction to Soft Skills** (TCS)
- **Faculty Development Program on Cloud Infrastructure** (BrainOVision) · **Getting Started with Amazon Web Services** (SkillUp)